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An efficient and recyclable Ni_2P–$Co_2P/ZrO_2/C$ nanofiber photocatalyst for the conversion of plastic waste into H_2 and valuable chemicals†

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The ubiquity of plastic waste has resulted in significant deleterious impacts on the environment and human health. Solar-driven photoreforming technology presents a facile and energy-efficient method for converting plastic waste into H_2 and valuable chemical compounds. In this study, we fabricated Ni_2P–$Co_2P/ZrO_2/C$ nanofibers (NFs) through a combination of electrospinning, carbonization, hydrothermal, and phosphorylation techniques. The Ni_2P–$Co_2P/ZrO_2/C$ composite NFs exhibit a coherent and homogeneous fibrous morphology, wherein Ni_2P–Co_2P nanowires are uniformly distributed and coated onto the surface of ZrO_2/C NFs. These nanofiber films demonstrate pronounced light absorption characteristics, showcasing their proficiency in efficiently capturing incident light. In an aqueous medium, polyethylene terephthalate (PET) can be effectively transformed into clean energy H_2, and valuable organic products using Ni_2P–$Co_2P/ZrO_2/C$ NFs as photocatalysts. Notably, the 24 hour rate of hydrogen production reached a significant value of 207.56 μmol H_2g sub^{-1}. The presented photocatalyst exhibits robust photocatalytic performance for at least four days under severe alkaline conditions. Furthermore, the catalyst is easy to isolate and recover from an aqueous solution after catalysis. In the context of photoconversion applied to authentic PET samples, which include polyester fibers and PET cups, the hydrogen production performance demonstrated by the catalyst is comparable to that achieved with pure PET particles. This congruence substantiates the practical viability of the system, affirming its applicability in real-world scenarios.

1. Introduction

A multitude of plastic articles derived from fossil resources function as efficacious implements in daily life.[1–3] The improper disposal of post-consumer plastics contaminates water bodies, endangering aquatic fauna, and posing a potential risk to human health and safety as it can enter the human food chain.[4–6] Various measures have been implemented to address the substantial negative impacts of plastic waste. Current global methods for managing plastic waste include landfilling, incineration, granulation-based regeneration, and pyrolysis.[7,8] However, these traditional approaches are ineffective and may lead to significant environmental pollution.[5] The current scope of plastic recycling is constrained, with most recycled plastic undergoing downcycling processes, resulting in lower-value items that are difficult to recycle further.[9] Consequently, the development of a mild and environmentally friendly novel technology for managing plastic waste has emerged as a pressing issue.

Photocatalytic technology, which utilizes solar energy for chemical oxidation reactions, presents several advantages, including high efficiency, environmental cleanliness, and cost-effectiveness.[10] This technology finds diverse applications, encompassing the degradation of organic pollutants, hydrogen production *via* water splitting, and the synthesis of organic compounds.[11–13] Furthermore, it holds promise for addressing the challenges in plastic waste management. In previous studies, Reisner *et al.* successfully synthesized carbon nitride/nickel phosphide (CN_X/Ni_2P) and CdS/CdO_X quantum dots (QDs) and employed them as photocatalysts for the efficient conversion of PET and PLA into carbonaceous fuels and H_2.[14,15] However, the toxicity and recyclability of the CdS/CdO_X system, and the low-rate limitation of the CN_X/Ni_2P catalyst restrict their practical application. Tajkia *et al.* employed visible light to decompose fragmented LDPE microplastics using visible light-induced plasmonic

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photocatalysts. These photocatalysts consisted of platinum nanoparticles deposited on zinc oxide (ZnO) nanorods (ZnO–Pt) and served as catalysts, resulting in the formation of low molecular weight compounds and surface deformities. It is worth noting that the use of a costly platinum (Pt) catalyst was a part of this process.[16] However, it is important to consider that powdered catalysts present challenges in terms of their separation from the reaction medium and may not be ideal for continuous processes. Additionally, noble metal-based catalysts, while demonstrating exceptional catalytic activity, are limited by their high cost and may not be suitable for all scenarios.[17]

In our previous research, we fabricated flexible $Ni_2P/TiO_2/C$ nanofiber films and employed them for photocatalytic PET degradation and hydrogen generation.[18] Carbon-based materials enhance nanoscale semiconductor separation and recycling through adsorption-photocatalysis synergy. Highly microporous carbon nanofibers exhibit substantial pore volume, extensive specific surface area, a uniform pore size distribution, and elevated adsorption capacity, making them valuable for environmental remediation.[19–21] However, the growth of Ni_2P nanoparticles on the surface of TiO_2/C nanofibers exhibits non-uniformity and incomplete coverage on certain fiber surfaces. This results in a reduced Ni_2P loading, affecting the catalytic efficiency. While Ni_2P-modified TiO_2/C nanofibers exhibit improved catalytic activity compared to TiO_2/C nanofibers, the overall efficiency remains relatively low. In contrast to TiO_2, zirconium dioxide (ZrO_2) possesses distinctive features as a semiconductor photocatalyst, including amphotericity, redox characteristics, abundant oxygen vacancies, and weak hydrophilicity.[22] When ZrO_2 is rendered with nanometer size and a unique morphology with a channel structure, it exhibits a large specific surface area, excellent adsorption capabilities, and a heightened capacity for strong ion exchange.[23] Moreover, besides demonstrating commendable catalytic performance, ZrO_2 serves as a versatile catalyst support capable of accommodating various catalytically active components.[24] ZrO_2 is a transition metal oxide with a wide bandgap (ranging from 3.3 eV to 5.1 eV) and exhibits significant photocatalytic activity under ultraviolet light (approximately 5% of the solar spectrum).[25] However, the high carrier recombination rate and low charge separation efficiency significantly impede the widespread application of zirconia. Given the surface chemical state and energy band structure, the incorporation of other semiconductors or co-catalysts with smaller band gaps, such as metal phosphides,[26,27] metal sulphides,[28–30] and carbides,[31] becomes crucial for semiconductor modification.

In addition, multiple metal phosphides have demonstrated superior photocatalytic activity compared to single-metal phosphide modifications, attributed to the synergistic effects arising from cooperative interactions among the constituent components.[32] Therefore, the pursuit of discovering an appropriate match for single-metal phosphide photocatalytic co-catalysts has great significance, given its potential to enhance catalytic performance and enable effective utilization in various applications. Inspired by this, herein, we synthesized Ni_2P–Co_2P nanowire-supported ZrO_2/C nanofibers (NFs) using a combination of the electrospinning technique and hydrothermal methods. Granular PET was chosen as the substrate to evaluate the photocatalytic activity of Ni_2P–$Co_2P/ZrO_2/C$ NFs for degrading plastic waste. The generation of H_2 and diminutive organic compounds was investigated using gas chromatography coupled with thermal conductivity detection (GC-TCD) and nuclear magnetic resonance spectroscopy (^{1}H NMR), and also the photocatalytic mechanism of the catalyst was investigated. It is demonstrated that under alkaline conditions, Ni_2P–$Co_2P/ZrO_2/C$ NFs possess the ability to produce both hydrogen and valuable chemicals from PET. Furthermore, the above system was utilized for the purpose of evaluating its performance in photoreforming of real-world plastic waste, namely polyester fibers and PET cups. Notably, the H_2 production activity of these materials was found to be comparable to that of PET particles, indicating their significant potential for practical implementation in the photoreforming method.

2. Experimental section

2.1. Materials

Cobalt nitrate ($Co(NO_3)_2{\cdot}6H_2O$), nickel nitrate ($Ni(NO_3)_2{\cdot}6H_2O$), and potassium hydroxide (KOH) were acquired from Shanghai Maclin Biochemical Co., LTD. Polyacrylonitrile (PAN, M_w = 150 000), urea (CH_4N_2O), zirconium acetylacetone ($Zr(acac)_4$), and sodium hypophosphite (NaH_2PO_2) were obtained from Sigma-Aldrich. *N*,*N*-Dimethylformamide (DMF, AR) was purchased from Hangzhou Gaojing Fine Chemical Co., LTD. All chemicals utilized in the study were of analytical grade and employed without undergoing additional purification steps. Throughout the entire experimental procedure, deionized water was employed as the solvent.

2.2. Preparation of the photocatalyst

The procedure for the fabrication of Ni_2P–$Co_2P/ZrO_2/C$ NFs is described in Scheme 1. In a typical process, 1.0 g of PAN was dissolved in 9.0 g of DMF solvent by magnetic stirring at room temperature for 12 h. Then 0.3 g of $Zr(acac)_4$ was added into the above solution under stirring to form a homogeneous solution. The prepared solution was then served as the precursor solution for electrospinning. During the spinning process, the solution was delivered at a flow rate of 0.85 mL h^{-1}, with an electric voltage of 18 kV and a distance of 15 cm between the spinning nozzle and the collector. The nanofibers were collected on the rotating collector, which was revolving at a speed of 255 rpm. After the electrospinning process, the collected fibers were peeled off from the collector and dried at 80 °C in a vacuum oven for 12 h. Afterwards, the fibers were stabilized at 320 °C for 60 min in air, with a heating rate of 1 °C min^{-1}, and then calcinated at 900 °C for 60 min under a N_2 atmosphere, with a heating rate of 5 °C min^{-1}. The ZrO_2/C NFs were obtained.

Subsequently, the ZrO_2/C NF film with a size of (2×2 cm^2) was subjected to three-step treatment involving acetone, diluted HCl, and deionized water, each performed three times, followed by thorough drying at 45 °C. The precursor material was prepared by a hydrothermal method. In a regular process, $Ni(NO_3)_2{\cdot}6H_2O$ (3 mmol), $Co(NO_3)_2{\cdot}6H_2O$ (3 mmol) and Urea

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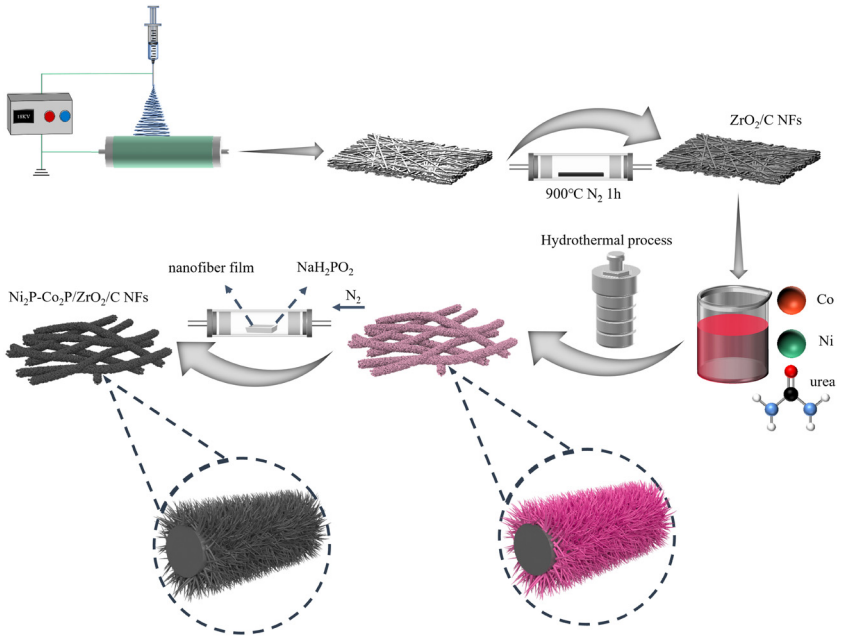

Scheme 1 The schematic illustration of the synthesis process of Ni_2P–Co_2P/ZrO_2/C NFs.

(6 mmol) were dissolved in 30 mL of distilled water and stirred to form a homogeneous solution. Then the solution and the as-cleaned ZrO_2/C NFs were transferred into a 50 mL Teflon-lined stainless autoclave.

The autoclave was sealed and placed in an oven, maintained at a temperature of 180 °C for 12 h, and then naturally cooled to room temperature. The nanofibers, loaded with the Ni–Co precursor, were removed, washed with deionized water, and dried at 60 °C. Following this, the nanofibers and a phosphidation agent, NaH_2PO_2 (in a mass ratio of 1 : 20), were placed at the upstream and downstream sides, respectively, of a horizontal porcelain crucible. This crucible was then positioned in a furnace, and the temperature was gradually increased to 350 °C at a heating rate of 2 °C min^{-1}, and maintained for 5 h under a flow of N_2 gas. As a result, Ni_2P–Co_2P/ZrO_2/C NFs were successfully prepared.

2.3. Characterization

A field-emission scanning electron microscope (FESEM, ZEISS ULTRA 55, Germany) and a high-resolution transmission electron microscope (HR-TEM; JEOL JEM2100, Japan) equipped with an energy-dispersive X-ray spectrometer (EDX; Super-X) were utilized to analyze the morphologies, elemental compositions and lattice fringes of the as-prepared samples. The phase composition was obtained using powder X-ray diffraction (XRD, Bruker D8 Discover, Germany) with Cu Kα radiation (λ = 1.5406 Å) in the 2θ range of 5–80°. X-ray photoelectron spectroscopy (XPS) measurements were recorded with an X-ray photoelectron spectrometer (K-Alpha Thermo Fisher Scientific USA). The binding energies of all samples were corrected with reference to the C 1s standard peak of 284.8 eV. Ultraviolet-visible (UV-vis) spectra were acquired using a Lambda 750 UV-vis-NIR spectrometer (Hitachi UH4150, Japan) equipped with an integrating sphere with fine $BaSO_4$ powder as the reference. The transfer and separation efficiencies of photogenerated electron–hole pairs at an excitation wavelength of 375 nm were examined utilizing a PG2000-Pro-EX photoluminescence spectroscopy spectrophotometer (FS5, Edinburgh Instruments). Photocurrent response and Electrochemical impedance spectra were recorded using an electrochemical workstation (Chenhua, CHI760E) equipped with a standard three-electrode cell configuration. In this configuration, the working electrode (WE) consisted of the catalyst grown on F-doped tin oxide (FTO), while a Pt wire and Ag/AgCl were employed as the counter and reference electrodes, respectively. The electrochemical characterization was carried out in a 0.5 M Na_2SO_4 aqueous solution.

2.4. Photocatalytic performance

Granular PET was selected as plastic waste to evaluate the photocatalytic activity of Ni_2P–Co_2P/ZrO_2/C NFs. In a typical experiment, PET particles were pretreated in KOH aqueous solution: 50 mg of PET was dissolved in 10 mL of 10 M KOH aqueous solution for 48 h at 60 °C with stirring at 400 rpm in the dark, to break up the granular PET. This process ensured enhanced contact between the catalyst and the degradation substrate, thereby resulting in improved photocatalytic efficiency. The pretreated PET solution was used for photocatalytic experiments (including incompletely dissolved polymer fragments). Then the photocatalyst weighing 200 mg and 10 mL of the pretreatment solution were introduced into a 45 mL quartz photoreactor, and the system was subjected to a 10 min N_2 purging at atmospheric pressure. The photoreactor was tightly sealed using a rubber stopper and positioned within a photochemical reaction device (YZ-GHX-A Y.Z. INSTRUMENT) equipped with a 500 watt mercury lamp. To analyze the evolving H_2 by the reaction, 400 μL of gaseous products were periodically

withdrawn from the test tube using a microliter syringe and analyzed using a gas chromatograph (GC-TCD, 1690T) equipped with a thermal conductivity detector and a molecular sieve column with N_2 as the carrier gas. To elucidate the influence of distinct conditions on the degradation efficacy of the photocatalytic system, a series of comparative experiments were devised, encompassing all pertinent factors, such as cocatalyst loading, alkali concentration, light intensity, and substrate treatment. Hydrogen production was chosen as the specified criterion for evaluating system performance.

Unless otherwise specified, all analytical measurements were conducted in triplicate, and the results are reported as the unweighted mean ± standard deviation (σ). The H_2 yield per unit weight of substrate (μmol H_2g sub^{-1}) and its corresponding standard deviation (σ) were calculated using eqn (1) and (2), respectively.

$$\text{Yield}\left(\mu\text{mol H}_2\text{g sub}^{-1}\right) = \frac{n_{\text{H}_2}}{n_{\text{substrate}}} \times 100\% \quad (1)$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})}{n - 1}} \quad (2)$$

where n is the number of repeated measurements, x is the value of a single measurement, and $\bar{x}$ is the unweighted mean of the measurements. σ was increased to 5% of $\bar{x}$ in the event that the calculated σ was below this threshold.

In order to characterize the degradation of PET, D_2O/NaOD was used instead of H_2O/KOH for pretreatment and the subsequent light reforming test. ^{1}H NMR spectra were recorded for quantitative analysis of PET light using a nuclear magnetic resonance spectrometer (NMR, AVANCEAV400 MHz). Changes in the monomers of each component before and after catalysis are recorded. The cycling experiments involved a total of 7 cycles with a cumulative duration of 168 hours and the duration of each cycle was 24 hours. The catalyst was initially separated from the suspension through centrifugation, followed by multiple washes employing DI water and alcohol. Then, the catalyst underwent a vacuum drying process at a temperature of 60 °C. Finally, the dried catalyst was weighed and subsequently employed for reuse. The catalyst after the last cycle was recovered and characterized for its structural stability.

3. Results and discussion

In order to elucidate the phase composition and microstructure of the nanofibers, they were characterized by FESEM. The FESEM image of ZrO_2/C nanofibers (NFs) is shown in Fig. 1(a) (after calcination at 900 °C). Obviously, the synthesized nanofibers were cylindrical with smooth surfaces. After hydrothermal treatment, it can be clearly observed in Fig. 1(b) that the surface of NFs is uniformly coated with many needle-like nanowires, and the fiber diameter increased significantly. Interestingly, after phosphidation as shown in Fig. 1(c), the fiber still maintained the intact fiber structure. Subsequent phosphorylation treatment yielded a perceptibly smoother fiber surface. The needle-like morphology was expected to be beneficial for facilitating the reflection of light among the fiber surfaces, thereby enhancing the sunlight absorption capacity and enhancing the photocatalytic performance.

Fig. 2 shows the XRD patterns of different samples. For ZrO_2/C NFs, the prominent diffraction peaks are located at 2θ values of 30.17°, 50.57° and 62.74°, and they are indexed to the (101), (200) and (202) planes of tetragonal ZrO_2, respectively, according to JCPDS #50-1089. The peaks at 35.16° and 60.12° reveal the presence of the (200) and (311) planes of cubic ZrO_2 according to JCPDS #27-0997, respectively. The distinctiveness and sharpness of the peaks in the diffraction pattern denote the high crystallinity of the synthesized nanocomposites. Additionally, the diffraction pattern reveals the coexistence of both tetragonal and cubic phases of ZrO_2 within the samples.[33] The absence of characteristic peaks at 2θ values of 24°, 28°, and 32° in the XRD pattern suggests the absence of monoclinic zirconia in the sample. This absence can be attributed to the

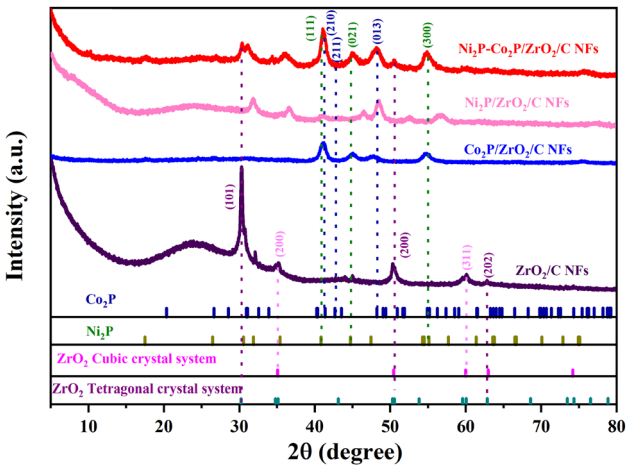

Fig. 2 XRD patterns of ZrO_2/C NFs, Co_2P/ZrO_2/C NFs, Ni_2P/ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs.

Fig. 1 SEM images of (a) ZrO_2/C NFs, (b) $Ni(OH)_2$–$Co(OH)_2/ZrO_2$/C NFs, and (c) Ni_2P–Co_2P/ZrO_2/C NFs.

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concurrent formation of ZrO_2 and carbon graphitization. Nevertheless, the carbon graphitization within the nanofibers impedes the migration of Zr and O species, consequently restraining the crystallization process of ZrO_2.[34,35] After hydrothermal treatment, discernible diffraction peaks corresponding to $Ni(OH)_2$ (JCPDS #14-0117) and $Co(OH)_2$ (JCPDS #51-17317) emerged on the surface of the nanofibers (Fig. S2, ESI†), signifying the generation of $Ni(OH)_2$ and $Co(OH)_2$ species on the ZrO_2/C NFs surface. Based on the above hydroxide precursors, the corresponding phosphides were synthesized through a phosphatization method. The characteristic peaks of Ni_2P/ZrO_2/C NFs and Co_2P/ZrO_2/C NFs match those of Ni_2P (JCPDS #03-0953) and Co_2P (JCPDS #74-0287), respectively. The characteristic peaks of Ni_2P–Co_2P/ZrO_2/C NFs are consistent with Co_2P (JCPDS #74-0287) and Ni_2P (JCPDS #03-0953), respectively. The phosphide product showed strong diffraction peaks at 40.4°, 44.7° and 54.2°, corresponding to the (111), (021) and (300) planes of Ni_2P, respectively. The characteristic peaks at 40.8°, 42.6°, and 48.2° are assigned to the (210), (211), and (013) crystal planes of Co_2P, respectively. These pieces of evidence support the simultaneous formation of both Co_2P and Ni_2P. Furthermore, the peaks at 30.2° and 50.2° correspond to tetragonal ZrO_2, which implies the presence of ZrO_2 in the substrate of Ni_2P–Co_2P/ZrO_2/C NFs.

In order to gain insight into the phase composition and microstructure of the NFs, it was further investigated by EDX and TEM. EDX spectra in Fig. S1a (ESI†) indicate that C and O elements are evenly distributed on the surface of carbon nanofibers. The elemental mapping (Fig. 3) displays the distribution of C, O, Zr, Co, Ni and P elements on the surface of nanofibers in the Ni_2P–Co_2P/ZrO_2/C NF sample, which is also confirmed by the EDX spectra in Fig. S1b (ESI†).

The TEM image reveals a distinctive configuration wherein nanowires enwrap the surface of the nanofiber, a finding that aligns coherently with SEM outcomes (Fig. 4(a)). Fig. 4(b) exhibits an HR-TEM image of Ni_2P–Co_2P/ZrO_2/C NFs. The images reveal distinct interplanar spaces of 0.28 nm and 0.29 nm, which correspond respectively to the (111) crystal plane of tetragonal ZrO_2 and the (101) crystal plane of cubic ZrO_2.[36] This observation provides further confirmation of the coexistence of two crystallographic phases of ZrO_2 within the carbon nanofibers. Furthermore, distinct interplanar spaces of 0.22 nm and 0.24 nm are observed on the nanofiber surface, and attributed to the (111) crystal plane of Ni_2P and the (210) crystal plane of Co_2P, respectively (Fig. 4(c)). This definitive identification provides additional evidence supporting the presence of Co_2P and Ni_2P species on the nanofiber surface.[15,37]

The XPS survey spectrum was employed to reveal the chemical state and composition of Ni_2P–Co_2P/ZrO_2/C NFs and ZrO_2/C NFs as shown in Fig. 5 and Fig. S3 (ESI†). The survey spectra (Fig. S3, ESI†) confirmed the main elements of Zr, C and O are presented on the surface of ZrO_2/C NFs. The full spectrum of Ni_2P–Co_2P/ZrO_2/C NFs is shown in Fig. 5(a). It can be clearly seen that the sample composed of C, O, Zr, Co, Ni and

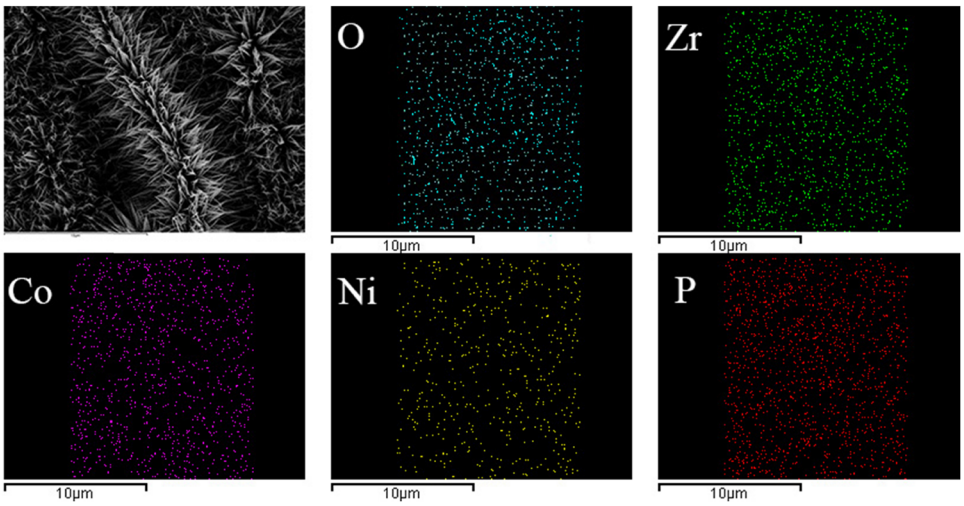

Fig. 3 The corresponding elemental mapping of the marked area in Ni_2P–Co_2P/ZrO_2/C NFs.

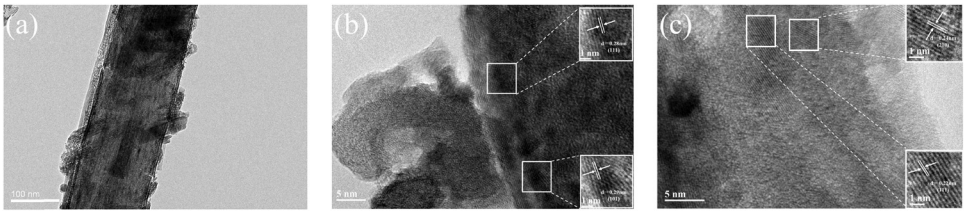

Fig. 4 TEM images of (a) Ni_2P–Co_2P/ZrO_2/C NFs; HR-TEM images of (b) and (c) Ni_2P–Co_2P/ZrO_2/C NFs.

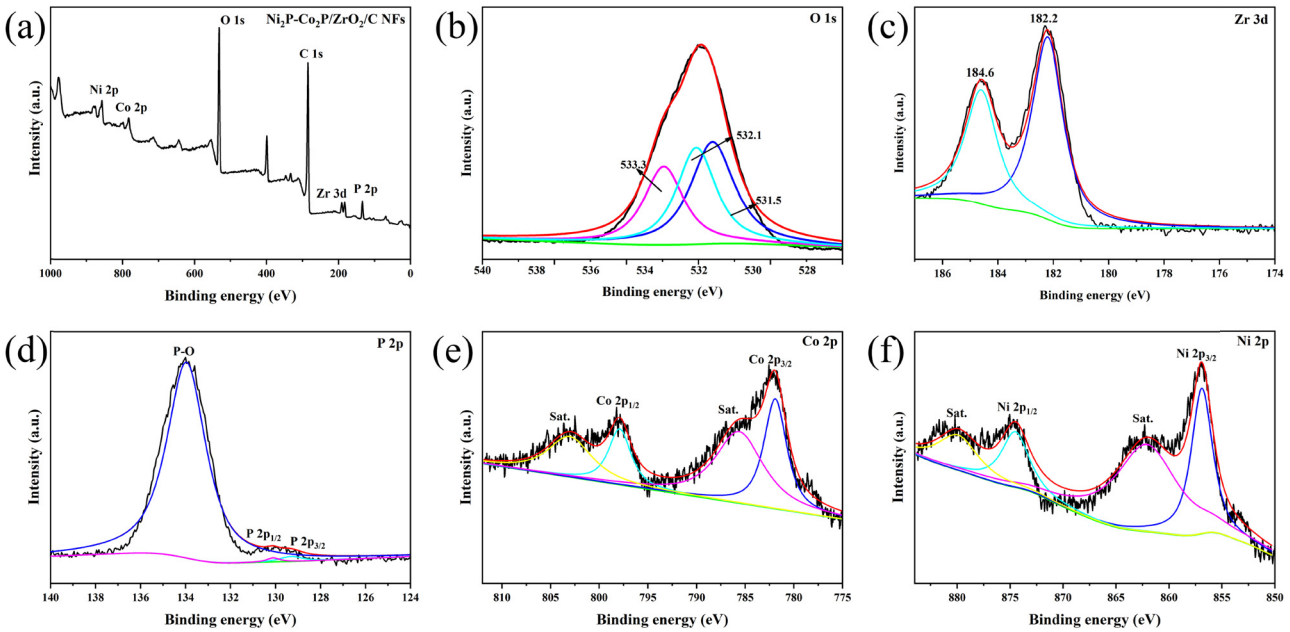

Fig. 5 XPS characterization of Ni_2P–$Co_2P/ZrO_2/C$ NFs: (a) survey spectrum, (b) high-resolution spectrum of O 1s region, (c) Zr 3d region, (d) P 2p region, (e) Co 2p region and (f) Ni 2p region.

P elements, which is consistent with the EDX result. The O 1s high-resolution XPS spectrum (Fig. 5(b)) shows three typical peaks located at 532.1 eV, 531.5 eV and 533.3 eV, which can be ascribed to the lattice oxygen of ZrO_2,[38] adsorbed oxygen,[39] surface carbonate or hydroxyl species (hydroxyl water absorbed on ZrO_2),[40] respectively. The high-resolution XPS spectrum of Zr 3d is shown in Fig. 5(c), the two peaks at 182.2 eV and 184.6 eV are assigned to Zr $3d_{5/2}$ and Zr $3d_{3/2}$, respectively. This corresponds to Zr^{4+} in ZrO_2.[41] In the P 2p spectrum (Fig. 5(d)), two peaks at 128 eV and 134.2 eV are attributed to the metal phosphide and phosphorus oxide species, assigned to P $2p_{3/2}$ and P $2p_{1/2}$, respectively.[31,42] The peak at 133.8 eV corresponds to the surface oxidation of metal phosphide caused by prolonged contact with air.[43] In the high-resolution Co 2p region (Fig. 5(e)), the spectrum can be deconvoluted to three spin–orbit doublets. The peak at 781.9 eV is attributed to Co $2p_{3/2}$,[44] corresponding to the cobalt oxide species (Co^{2+}) caused by surface oxidation of Co–P in air. While the peak at 786.1 eV belongs to the satellite of the Co $2p_{3/2}$ peak.[45,46] The Ni 2p XPS spectrum is shown in Fig. 5(f), and the peak at 856.5 eV is assigned to Ni $2p_{3/2}$, while the peak at 861.9 eV belongs to the satellite of the Ni $2p_{3/2}$ peak.[47] The peak at 873.6 eV could be attributed to the Ni $2p_{1/2}$ spin–orbital splitting of Ni^{2+}.[48] All the above characteristics confirm the successful conversion of Ni–Co precursors to Ni_2P–Co_2P on ZrO_2/C NFs.

The light absorption ability is a key for the performance of photocatalysts, Thus, an investigation into the light absorption ability of Ni_2P–$Co_2P/ZrO_2/C$ NFs and other relevant samples was conducted by ultraviolet-visible (UV-vis) diffuse reflectance spectroscopy. As shown in Fig. 6, the pristine ZrO_2 exclusively manifests ultraviolet absorption, with an absorption edge at around 220 nm. This absorption phenomenon is chiefly

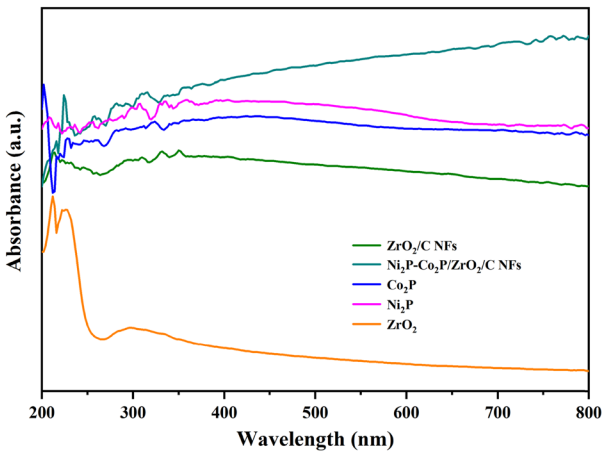

Fig. 6 UV-Vis absorption spectra of Co_2P, Ni_2P, ZrO_2, Ni_2P–$Co_2P/ZrO_2/C$ NFs, and ZrO_2/C NFs.

attributed to transitions between the valence band and the conduction band, signifying the existence of oxygen vacancies within the crystalline lattice of ZrO_2.[49,50] The absorption peak at 300 nm is ascribed to the excitation of valence band electrons from O^{2-} to the conduction band of Zr^{4+}.[51] It is noteworthy that Ni_2P and Co_2P both exhibit strong optical absorption throughout the entire UV-visible range, which could be attributed to their black color.[52] When employing carbon nanofiber films as carriers, the absorption of ZrO_2/C composite nanofiber films is significantly enhanced in the range of 400–800 nm. This is attributed to the ability of the carbon fiber substrate to reduce light reflection.[53] The absorbance of the $Ni_2P/Co_2P/ZrO_2/C$ NFs is higher than that of other samples across the entire optical spectrum. This is attributed to the enhanced light absorption

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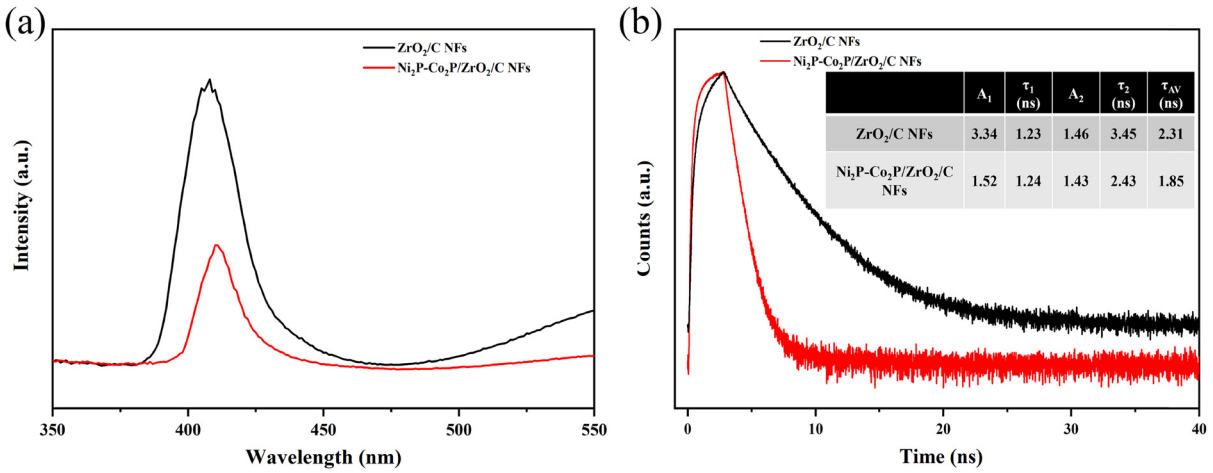
performance of the sample due to the loading of the cocatalyst. The synergistic effect between Ni_2P and Co_2P enhances the absorption of light.[54]

The PL spectra of ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs effectively elucidate the enhancement effect resulting from this modification. In Fig. 7(a), the PL spectrum of ZrO_2/C NFs reveals the presence of a relatively broad emission peak within the wavelength range of 375–500 nm. This emission peak is attributed to the recombination of photoexcited electrons and holes within ZrO_2/C NFs, resulting in the release of energy in the form of fluorescence. The introduction of cocatalysts onto Ni_2P–Co_2P/ZrO_2/C NFs leads to significant reduction in PL. This is evident from the significant reduction in the emission peak, indicating the effective mitigation of photoexcited charge and hole recombination. Besides, Ni_2P–Co_2P/ZrO_2/C NFs exhibit a red-shift in absorption at longer wavelengths (400–425 nm), thereby indicating enhanced utilization of light in the photonic regime. The investigation of the augmented separation of photoexcited electrons and holes was conducted through analysis of the photocurrent transient response. As shown in Fig. 7(b), the average lifetime of ZrO_2/C NFs is approximately 2.31 ns, whereas the average lifetime of Ni_2P–Co_2P/ZrO_2/C NFs is approximately 1.85 ns (inset Fig. 7(b)). The TRPL results clearly demonstrate that the effective loading of Ni_2P–Co_2P enhances charge transfer and separation.[55]

The charge separation and transfer efficiency in Ni_2P–Co_2P/ZrO_2/C NFs were subjected to further investigation through electrochemical impedance spectroscopy (EIS) and photocurrent transient response. In Fig. 8(a), the photocurrent of Ni_2P–Co_2P/ZrO_2/C NFs is 0.2 μA cm^{-2}, indicating an enhancement of approximately 4.0 times compared to ZrO_2/C NFs (0.05 μA cm^{-2}). The enhanced charge separation behavior was also testified by EIS technology. From Fig. 8(b), Ni_2P–Co_2P/ZrO_2/C NFs display a smaller Nyquist plots semicircle than pristine ZrO_2/C NFs, revealing that Ni_2P–Co_2P/ZrO_2/C NFs possess a favorable interfacial electron transfer capability and lower resistance. Based



	A_1	τ_1 (ns)	A_2	τ_2 (ns)	τ_{AV} (ns)
ZrO_2/C NFs	3.34	1.23	1.46	3.45	2.31
Ni_2P-Co_2P/ZrO_2/C NFs	1.52	1.24	1.43	2.43	1.85

Fig. 7 (a) PL spectra of ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs, (b) TRPL spectra of ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs. The inset image of (b) shows the calculated results of the carrier lifetime of ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs.

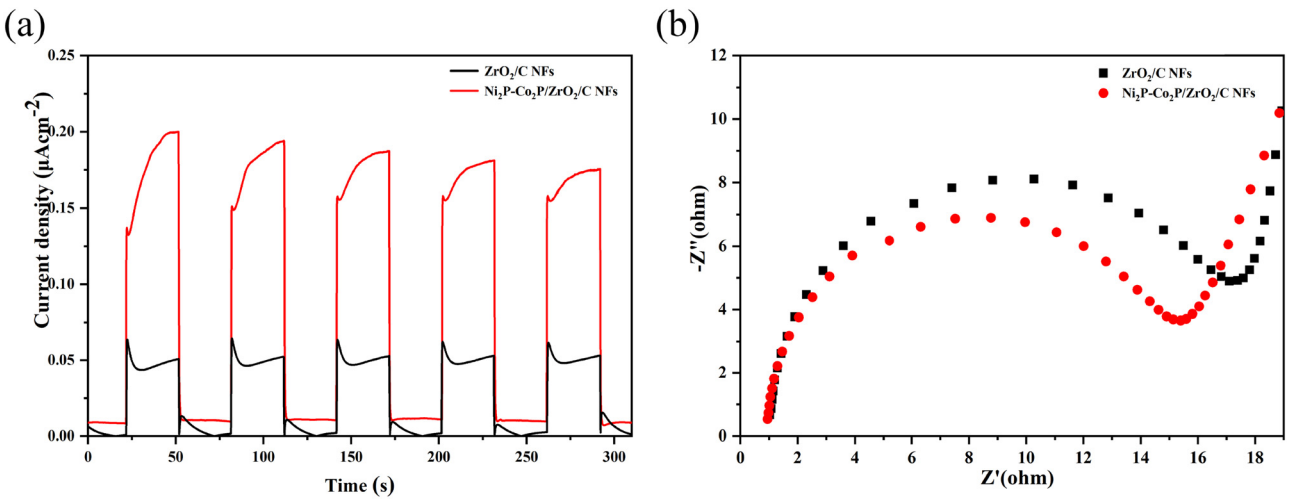


Fig. 8 (a) photocurrent transient response and (b) EIS of ZrO_2/C NFs and Ni_2P–Co_2P/ZrO_2/C NFs.

on the aforementioned photoelectric properties, the incorporation of cocatalysts in the form of Ni_2P–$Co_2P/ZrO_2/C$ NFs promotes the efficient separation of photoexcited charges. It is anticipated that Ni_2P–$Co_2P/ZrO_2/C$ NFs exhibit superior photocatalytic performance compared to ZrO_2/C NFs.

3.1. Photocatalytic activity

PET as a typical polymer-based plastic was employed to investigate the degradation performance of photocatalysts. We observed that hydrogen gas was generated within the same catalytic time when the pretreated solution was used as the substrate. This suggests that the pretreatment of PET, which alters the polymer properties and renders it more susceptible to degradation by active species, plays a significant role in facilitating subsequent H_2 production.[56] In the absence of a photocatalyst, illumination, substrate, or pretreatment, the detection of H_2 was notably absent, as indicated in Fig. 9 and Table S1 (ESI†). Furthermore, in the absence of PET as the degradation substrate, the photocatalytic system demonstrated no observable generation of H_2, as evidenced by Fig. 9(a) and Table S1 (ESI†). This can be attributed to the ease with which the carriers recombine in the absence of a hole scavenger, which in turn impedes the completion of the redox reaction. Furthermore, the hydrolysis of PET necessitates a considerable energy input, which is extremely thermodynamically challenging. In the presence of PET, it serves as a sacrificial reagent, participating in an effective combination with protons (h^+), thus mitigating carrier recombination and ultimately facilitating the efficient liberation of H_2. Moreover, it was discovered that the concentration of KOH serves a crucial role in H_2 evolution and the pretreatment of PET exhibited the highest activity in H_2 production (207.56 μmol H_2g sub^{-1}) from photoreforming of 24 hours at an alkalinity of 10 M (Fig. 9(b) and Table S2, ESI†). This phenomenon can be attributed to the accelerated hydrolysis of PET into its constituent monomers due to the elevated concentration of KOH.[57] However, as the concentration of KOH continues to increase, the increase in hydrogen production is not significant. This may be due to the transformation of the Ni_2P–Co_2P component into $Ni(OH)_2$ and $Co(OH)_2$ in a highly alkaline environment.[58] In order to reduce the costs and mitigate the corrosive nature of the system, all subsequent experiments were conducted in a 10 M KOH solution. The appropriate loading of a cocatalyst represents a pivotal factor in the enhancement of the photocatalytic efficiency exhibited by the sample. In the context of identical circumstances, it can be observed that the employment of Ni_2P–$Co_2P/ZrO_2/C$ NFs as a catalyst in the system results in significantly elevated levels of H_2 production in comparison to the use (34.47 μmol H_2g sub^{-1}) of ZrO_2/C NFs as the catalyst (Fig. 9(c) and Table S1, ESI†). The H_2 production rates observed for $Ni_2P/ZrO_2/C$ NFs and $Co_2P/ZrO_2/C$ NFs are markedly greater than those of ZrO_2/C NFs, yet comparatively lower than that of the Ni_2P–$Co_2P/ZrO_2/C$ NFs. This provides evidence that the incorporation of nanofibers containing Ni–Co phosphide yields a significant enhancement in photocatalytic efficiency. This is due to the prodigious and remarkable electron-trapping ability of Ni_2P–Co_2P, along with the synergistic effect of well-established contacts between the nanofibers and the catalyst. Thus expedite the expeditious transfer of electrons to Ni_2P–Co_2P, all of which culminate in the quelling of carrier recombination and the relentless promotion of hydrogen production.

Furthermore, apart from the production of hydrogen in the reduction half-reaction, the complete catalytic system additionally yields organic products of considerable value in the oxidation half-reaction.[15,59] The identification and characterization of these organic products are achieved through ^{1}H NMR spectroscopy, as illustrated in Fig. 10. After pre-processing in KOH solution, PET undergoes hydrolysis to terephthalate (TPA, a) and ethylene glycol (Eg, b), these substances were also found in the reaction system after catalysis (Fig. 10(a)). Photocatalytic experiments were carried out using the hydrolysis products of PET (Ethylene glycol and TPA) as the substrate, and the results are presented in Table S3 (ESI†). In the photocatalysis process, TPA was not oxidized (no H_2 production). This may be attributed to the challenging catalysis of the aromatic portion. Since TPA is sparingly soluble under most aqueous conditions, it is possible to separate and reuse it in PET synthesis. Catalytic reactions using pure EG as the substrate exhibit higher hydrogen generation activity (69.32 ± 4.57 μmol H_2g sub^{-1}, as shown in Table S3, ESI†). Its high reactivity exhibited by ethylene glycol renders it susceptible to oxidative degradation with a relatively greater ease.[15] The photoreforming process results in the generation of three carboxylate products, namely, glycolate (c), formate (d), and acetate (e). These products have been successfully identified by ^{1}H NMR spectroscopy (Fig. 10(b)), and similar

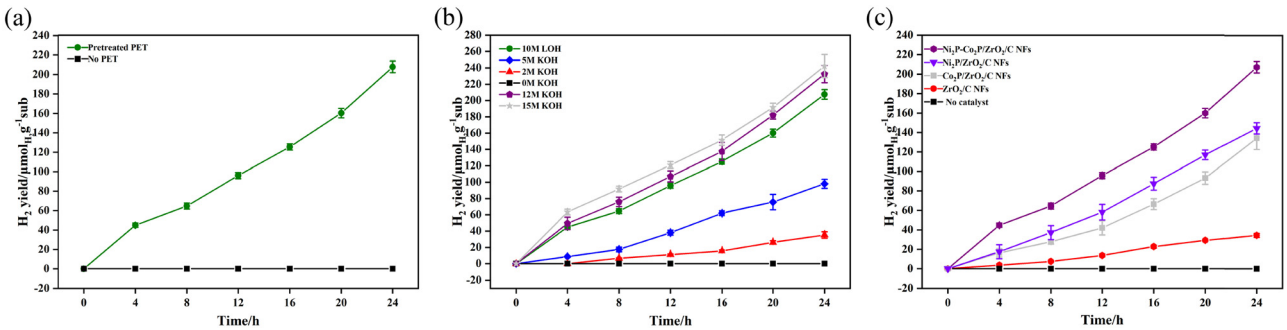


Fig. 9 Photocatalysis of PET with Ni_2P–$Co_2P/ZrO_2/C$ NFs, (a) pretreated PET and no PET, (b) KOH concentration for photoreforming, and (c) with Ni_2P–Co_2P loading, with Ni_2P loading, with Co_2P loading, without a co-catalyst loading and no catalyst.

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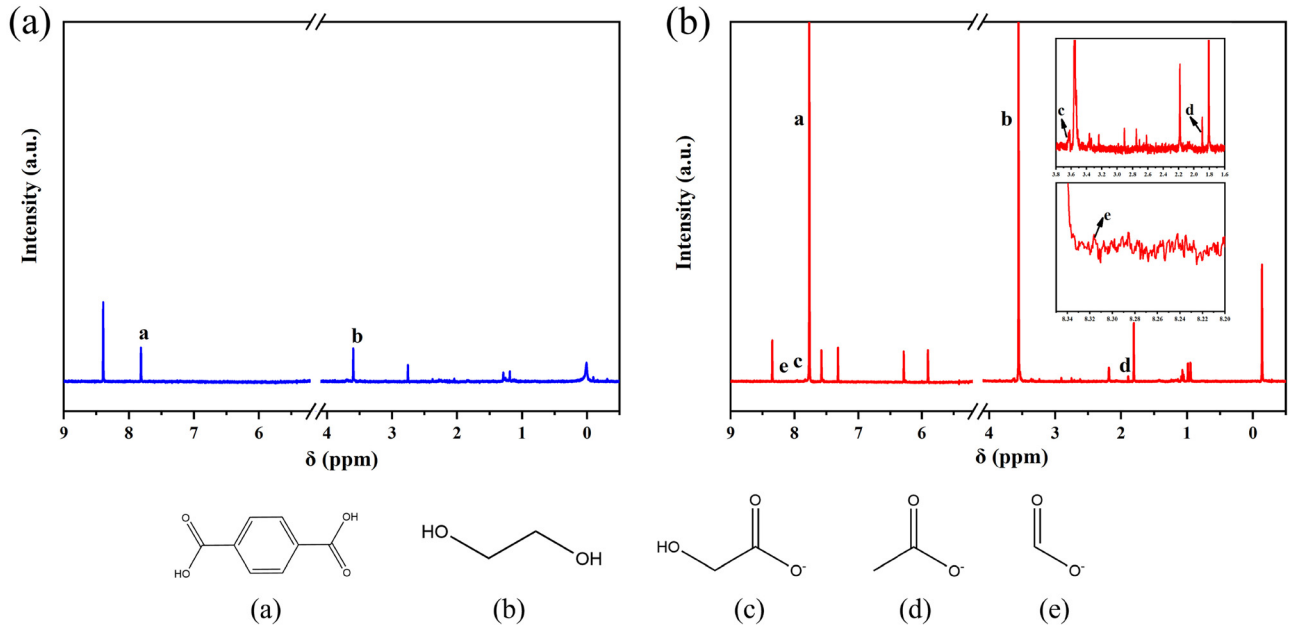


Fig. 10 ^{1}H NMR spectra of PET (a) after pre-treatment and (b) after photocatalysis. Insets show zoomed-out views of the spectra. The chemical structure formula corresponds to the peak assignment.

PET oxidation products have been previously reported,[60] further corroborating the consistency and relevance of the observed findings in the context of PET degradation.

According to antecedent research, $^{\bullet}O_2^-$, $^1O_2$, h^+ and $^{\bullet}OH$ are the principal dynamic constituents involved in the mechanism of photocatalytic degradation.[61] In order to elucidate the role of these species in the photocatalytic degradation of PET over the Ni_2P–$Co_2P/ZrO_2/C$ NF photocatalyst, a series of trapping experiments were conducted. Four different scavengers were employed to detect the involved species, including BQ, AO, L-histidine and IPA for capturing $^{\bullet}O_2^-$, h^+, $^1O_2$ and $^{\bullet}OH$, respectively (Fig. 11(a)). Under the conditions where the h^+ scavenger (AO) and the $^{\bullet}O_2^-$ scavenger (BQ) were present. The generation of hydrogen was inconsequential and the hydrogen production activity of the catalytic system in the presence of the $^1O_2$ scavenger (L-histidine) was significantly inhibited, but hydrogen was still produced in the reaction system. However, in the system where the $^{\bullet}OH$ scavenger (IPA) is present, there was a negligible reduction in the amount of hydrogen production. Obviously, the active constituents implicated play an important role in the process of photocatalytic degradation of PET. The primary catalytically active components within the system are h^+ and $^{\bullet}O_2^-$, whereas $^1O_2$ displays a weaker impact than h^+ and $^{\bullet}O_2^-$, and $^{\bullet}OH$ evinces no discernible influence on the catalytic process.

The determination of the bandgap of ZrO_2, Co_2P, and Ni_2P were conducted based on UV-visible diffuse reflectance

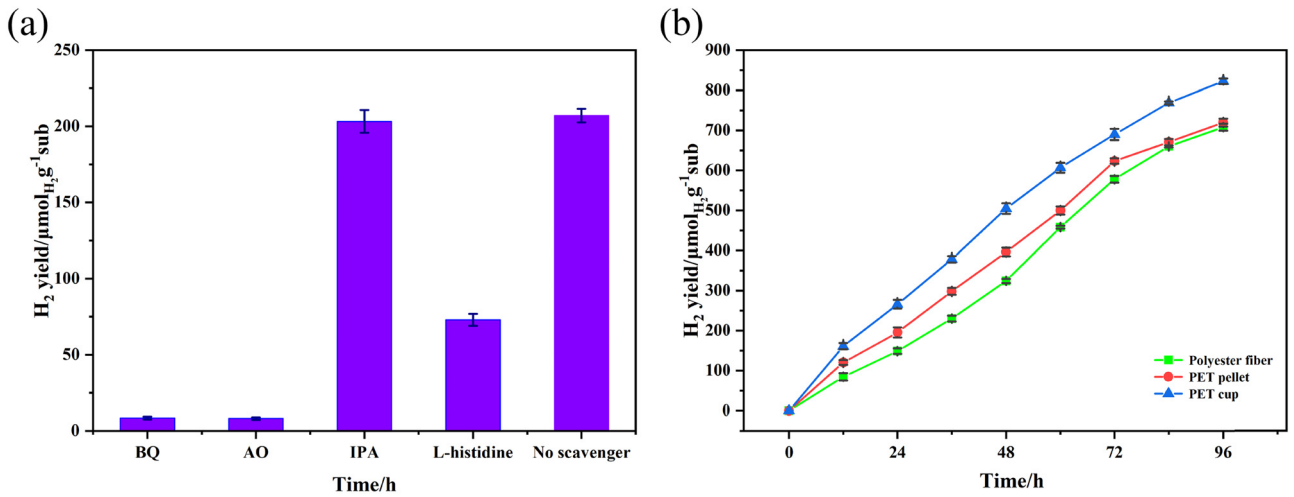


Fig. 11 (a) Photocatalysis of PET with Ni_2P–$Co_2P/ZrO_2/C$ NFs in the presence of different scavengers, and (b) Photoreforming of nonrecyclable plastic waste.

spectroscopy (DRS) analysis, as illustrated in Fig. S4 (ESI†). The calculated values for the bandgap were found to be 4.85 eV, 1.65 eV, and 1.65 eV, respectively. The XPS valence-band spectra implied that the valence band (VB) potential of ZrO_2, Co_2P and Ni_2P were 2.27 eV, 1.12 eV and 1.39 eV, respectively (Fig. S5, ESI†). Thus, the conduction band (CB) positions of ZrO_2, Co_2P and Ni_2P were calculated to be −2.58 eV, −0.53 eV and −0.26 eV, respectively, using eqn (3).

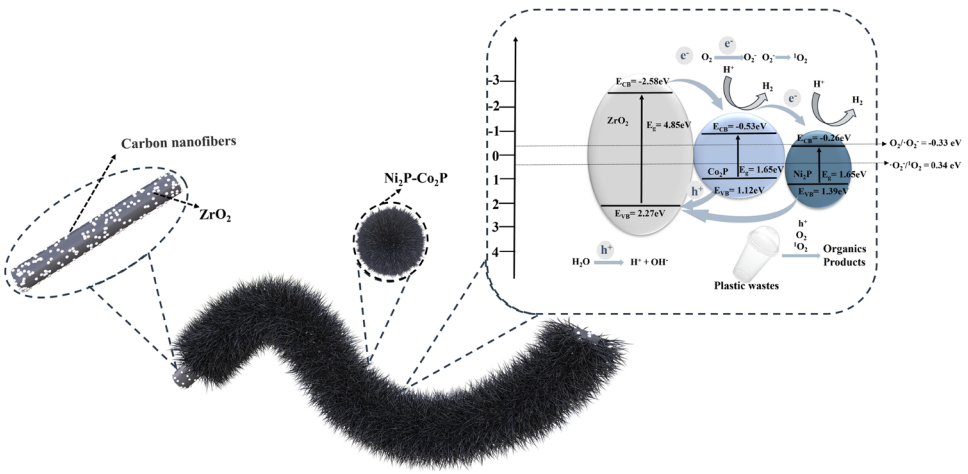
$$E_{VB} = E_{CB} + E_g \quad (3)$$

Fig. S6 (ESI†) depicts the Mott–Schottky plots of various samples, where the intercept of the linear portion of the Mott–Schottky curve on the *x*-axis precisely represents the flat-band potential (E_{fb}) of the samples. For all samples, the plots obtained at a frequency of 1000 Hz display a negative slope intersecting the *x*-axis. This implies that the prepared ZrO_2, Co_2P, and Ni_2P exhibit characteristics of n-type semiconductors, and an E_{fb} of ∼−0.07, −0.67 and −0.54 eV *versus* Ag/AgCl, respectively. In comparison to the reversible hydrogen electrode, the potential of Ag/AgCl can be approximated using eqn (4):

$$E_{fb} = \text{Ag/AgCl} + 0.059 \times \text{pH} + 0.197\ \text{V}. \quad (4)$$

In Scheme 2, we delineate the charge migration mechanism between Ni_2P–Co_2P and ZrO_2. Co_2P and Ni_2P, serving as reducing semiconductor catalysts, exhibit relatively small work functions, yet their Fermi levels are comparatively higher.[62] As for ZrO_2, it possesses a relatively high work function as an oxidizing semiconductor catalyst, along with a lower Fermi level.[63] This has been reported in previous reports. Therefore, electrons will transfer from Co_2P and Ni_2P, which have higher Fermi levels (E_f) to ZrO_2, causing the band structure to bend downward until the Fermi levels of both phases become equal.[44] In this scenario, photogenerated electrons and holes can swiftly undergo separation. In the presence of sunlight, the catalyst absorbs light energy, resulting in the excitation of electrons from the valence band to the conduction band and the generation of holes in the valence band. The electrons excited by light and transitioning from the VB will accumulate in the CB of ZrO_2. As electrons continue to accumulate, they can transfer to Co_2P and Ni_2P. The energy bands of Ni_2P and Co_2P are well-aligned, allowing electrons to directly transfer between different metal phosphides. Electrons on Ni_2P and Co_2P can quickly reduce H^+ to H_2, while some electrons react with oxygen to produce superoxide radicals, leading to the degradation of PET. h^+ acts directly as the active substance in the PET decomposition process. According to the experimental results, $^1O_2$ also participated in the photocatalytic process. $^1O_2$ was probably produced *via* the further oxidation of $^{\bullet}O_2^-$.[56] Experimental these species facilitated the degradation process and thus augmented the photocatalytic activity of Ni_2P–Co_2P/ZrO_2/C NF composites.

Having acquired an in-depth understanding of the photocatalytic performance and inherent photocatalytic mechanism of Ni_2P–Co_2P/ZrO_2/C nanofibers (NFs), a series of cyclic photocatalytic degradation experiments was conducted with the purpose of assessing the stability of the catalyst. These experiments aimed to assess the long-term stability and efficiency of the system through repeated cycles of photocatalytic activity. As shown in Fig. S7 (ESI†), the hydrogen production activity exhibited a notable decline starting from the fourth cycle compared to the initial cycle. Nevertheless, it is noteworthy that the nanofibers (NFs) retained their catalytic activity even after undergoing seven consecutive cycles of photocatalytic activity. To gain more insights, we compared the SEM and XRD patterns of Ni_2P–Co_2P/ZrO_2/C NFs before and after the photocatalytic degradation reaction (Fig. S8, ESI†). It was observed that after the photocatalytic experiment, the needle-shaped Ni_2P–Co_2P which was previously loaded onto carbon nanofiber films (NFs) has undergone complete disappearance. The fiber surface became rough, while the original fiber shape remained unaltered, which shows that the metal phosphide undergoes partial corrosion. Additionally, a decrease in the loading of the co-catalyst was observed in specific regions of the fiber surface, leading to the manifestation of a bare fiber state (Fig. S8a, ESI†).[26,64] The characterization of the catalyzed Ni_2P–Co_2P/ZrO_2/C NF catalyst by XRD further verifies the



Scheme 2 Mechanism diagram of the polymer photocatalysis with a Ni_2P–Co_2P/ZrO_2/C NF photocatalyst in alkaline aqueous solution.

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reason for this morphology (Fig. S8b, ESI†). It confirms that Ni_2P–Co_2P remains on the carbon nanofiber (NF) film surface. However, the intensity of the corresponding peak was observed to be significantly lower in comparison to the sample prior to the catalytic reaction, presumably due to the loss of surface PO_X species.[10] Moreover, subsequent to the reaction, distinct derivative peaks emerged on the catalyst, corresponding to cobalt hydroxide and nickel hydroxide, respectively. These findings suggest that under conditions of strong alkalinity in the experimental environment, the metal phosphide species were substituted by metal hydroxide species, leading to a higher presence of Ni–OH and Co–OH acting as a cocatalyst in subsequent cycling experiments, which is consistent with the literature.[65]

Finally, we evaluated the real-world effectiveness of the Ni_2P–Co_2P/ZrO_2/C NFs system by performing long-term photocatalysis on PET fibers, PET particles, and PET cups for food packaging (Fig. 11(b) and Table S4, ESI†). The essentiality of testing real-world plastic specimens lies in their tendency to harbor supplementary components, such as fillers, crosslinkers, or antioxidants, which could potentially impede the process of catalytic degradation. The present study ensures the conformity of various parameters, encompassing the concentration of the catalyst, the concentration of the substrate, and the pretreatment method, to maintain consistency with the photocatalytic circumstances of PET particles. Fig. 11(b) shows continuous H_2 generation from the PET cups, PET particles, and PET fibers over the course of 96 h, with an activity of up to yields of 822.13, 719.45, and 708.06 μmol H_2g sub^{-1} respectively. Initially, the production of hydrogen in PET fibers exhibited a comparatively reduced level, which is likely attributable to the inclusion of supplementary crosslinking agents and a decrease in solubility. The hydrogen production efficiency of PET fibers increases over time as more surface area is exposed to hydrolysis. The PET cups exhibited the highest hydrogen production activity, plausibly due to the relatively reduced contact area of granular PET with KOH aqueous solution and its diminished solubility, as opposed to the flake PET cup which presents a larger surface area for hydrolysis to occur. The foregoing statement presents an initial manifestation of the actual catalytic deterioration of plastic waste and concomitant hydrogen generation in practical settings.

4. Conclusions

In this work, we successfully fabricated Co_2P–Ni_2P modified ZrO_2/C nanofibers through a sequential process involving electrospinning, carbonization and phosphating, and employed them for the photocatalytic reforming of plastic waste. The synergistic effect of incorporating Ni_2P–Co_2P as a cocatalyst alongside ZrO_2 was demonstrated to effectively facilitate charge separation, ultimately leading to a notable improvement in catalytic efficiency. The Ni_2P–Co_2P/ZrO_2/C NF photocatalyst effectively produced hydrogen utilizing pretreated PET waste as the primary feedstock. Within a 24 hour duration, the Ni_2P–Co_2P/ZrO_2/C NF photocatalyst achieved a remarkable hydrogen yield of 207.56 μmol H_2g sub^{-1}, much higher than that (34.47 μmol H_2g sub^{-1}) of ZrO_2/C NFs as a photocatalyst. The composite photocatalyst also demonstrates the capability to generate other valuable organic substances *via* the oxidation half-reaction, consequently enhancing the overall process value of the system. Moreover, it exhibited remarkable ability to catalyze the degradation of PET cups and fibers, with the hydrogen production rates reaching 822.13 and 708.06 μmol H_2g sub^{-1}, respectively. These PET materials serve as genuine polymer specimens commonly utilized in practical applications, thereby exemplifying their practical utility. The nanofibrous substrate demonstrates outstanding stability in a robust alkaline environment owing to the safeguarding provided by the carbon shell, thereby facilitating its facile recyclability. Future research endeavors should focus on the refinement of the pretreatment technique, while simultaneously enhancing the light absorption range of the catalyst, as a means of advancing the ecological decomposition of plastic waste.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgements

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